Human-in-the-Loop Visual Analytics Dashboard for Image Classification and Description Editing

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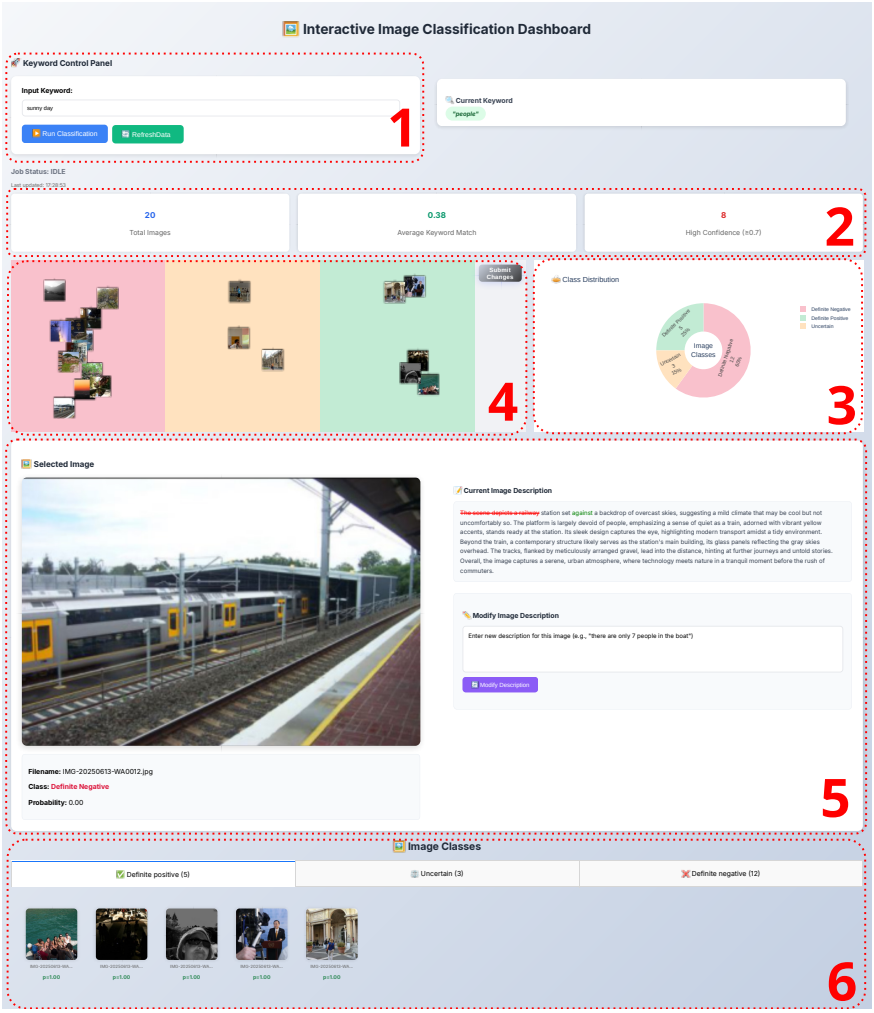

Fig. 1: This dashboard lets users classify images based on a given keyword in the *job control panel* [1], using both the images and their descriptions. After classification, users can review and edit image descriptions in two ways. The first is through a drag-and-drop Cytoscape interface [4], where images can be moved to indicate whether they contain the keyword. The second is by selecting an image via the "Classes" tab in *the image grid* [6], which brings up its details and an input box to suggest changes to the description[5]. The dashboard also provides dataset analytics, *the statistical overview* [2], including the number of images, the count of confidently classified images, and a pie chart[3] showing the distribution across three classes.

Index Terms—Foundation models, large language models (LLM), multimedia analytics, Semantic Interaction (SI), Text-to-image retrieval, vision-language models (VLM)

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ABSTRACT

We present a mixed-initiative visualization interface that enables users to dynamically categorise and explore image collections using natural language queries and semantic manipulation. Leveraging state-of-the-art vision-language models (VLMs) and large language models (LLMs), our system provides categorisation feedback for each image in response to user-defined keywords. It allows for flexible human-in-the-loop interaction through novel drag-and-drop adjustments, based on

ImageSI and IC|TC, and direct prompting to change image descriptions, supporting both subjective exploration and precise editing. Built on the Dash framework with a customized Cytoscape interface, that shows real-time updates for enhanced user understanding and control. We evaluate the system on a sample of the YFCC100M dataset, demonstrating its capacity to align model behavior with human intent while facilitating insight through mixed-level interaction. This work contributes to the emerging paradigm of foundation model-driven multimedia analytics and human-AI teaming.

1 Introduction

As image collections continue to grow in size, tools that enable users to efficiently retrieve specific collections and interpret images based on personalized criteria have become increasingly important. Traditional image retrieval methods often lack flexibility when users wish to query images based on highly specific or personal criteria. They typically rely on static embeddings and rigid search interfaces that fail to reflect the subtleties of user intent.

In our project, we address this gap by introducing a system that allows users to retrieve individual images based on natural language queries and receive image probability estimates for every image, indicating the relevance to the query (e.g., "sunny weather" or "outdoor activity"). Rather than grouping images into static groups, the system provides dynamic feedback on each image, enabling more flexible and transparent exploration of the image collection.

Our interface is inspired by prior work on interactive image clustering, such as IC|TC [6], and semantic interaction frameworks like ImageSI [8]. Although, we no longer use these methods directly to redefine and improve clusters. Instead, we adopt the idea of enabling users to edit or refine image descriptions through simple interactions, such as drag-and-drop, to better understand and adjust the system's interpretation of visual content.

This paper presents the design and implementation of our retrieval-based system, demonstrates its capabilities across different use cases, and discusses how providing probability-based feedback and interactive editing helps achieve greater insight into both the image collection and the underlying VLMs.
Our main contributions are:

1. A novel interaction technique that integrates semantic interaction and text-to-image retrieval. Inspired by ImageSI [8] and IC|TC [6], our technique allows users to express feedback by dragging and dropping images within a 2D projection space. If activated, the system updates the image descriptions in natural language, enabling the semantic embedding to align with user intent.

2. An intelligent, mixed-initiative visualization interface that combines direct manipulation and natural language input. Our dashboard supports user-driven refinement of image semantics through two mechanisms: spatial manipulation (drag-and-drop) and text-based editing of image descriptions. The system offers real-time feedback by updating classification outcomes and generating revised descriptions, thus facilitating iterative model alignment and interpretability.

2 Related Work

Our system draws on advances in interactive text-to-image retrieval, foundation models (both VLMs and LLMs), and human-in-the-loop interaction. It builds on the idea that users should be able to query image collections flexibly and refine system behaviour based on subjective needs.

2.1 Text-to-image retrieval

Text-to-image retrieval, a task focused on locating target images in an image database that correspond to an input text query, has seen significant advancements thanks to the development of VLMs [7]. VLMs such as CLIP, ALIGN, BLIP-2, SigLIP and SigLIP-2 that have powerful zero-shot capabilities have now become the default method for openset text-to-image retrieval [18]. Most of these models have failure cases where small, but semantically important objects in an image are not properly understood by the model. We try to inspect this, by giving users the possibility to evaluate the results of their text-to-image retrieval.

2.2 Foundation Models for Vision–Language Interaction

Foundation models have advanced at a remarkable pace in recent years. Combined with instruction tuning, they can now be flexibly adapted to a wide range of downstream tasks [12]. In particular, LLMs demonstrate impressive performance across diverse natural language processing tasks, such as text summarization. Moreover, Radford et al. [11] showed that LLMs can successfully address computer vision problems without a direct linguistic component, an insight that also motivates our work.

Vision–language models (VLMs) have likewise progressed rapidly. They can now generate image descriptions that reflect user-defined prompts with increasing accuracy and nuance. For example, IC|TC [6] demonstrated that users can specify arbitrary criteria (e.g., "group by mood" or "group by activity"), enabling highly personalized results. Building on this idea, our system does not output fixed clusters; instead, it provides probability scores for every image, that indicate how well each image matches user-defined keywords. This approach optimizes detailed exploration while avoiding rigid grouping.

Beyond pure text-to-cluster pipelines, metadata-driven explanations and multimodal constraints play a key role in enhancing interpretability and control. Mozolewski et al. [9] introduce an algorithm that not only clusters images but also generates textual summaries of each cluster using associated metadata, and even lets users rename clusters to improve explainability and guide the selection of cluster counts. We build on this insight by making the user able to adjust textual descriptions, on which the text-to-image retrieval is based

Additionally, Hirota et al. [4] demonstrate that integrating photo metadata (e.g., timestamps, GPS) with low-level image features via constraint-based clustering yields more semantically coherent groupings in image retrieval scenarios. Although our focus remains on text-driven clustering, their work underscores the value of fusing multiple modalities to refine cluster boundaries. Accordingly, our pipeline supports optional metadata and visual-feature constraints, users can specify criteria like "images that are taken at night" or "images that are taken in Europe", further improving both the meaningfulness and explainability of the resulting classification.

Together, IC|TC's natural-language interface, metadata-powered explanations, and multimodal constraint mechanisms form a cohesive text-based classification management component that empowers users to define, interpret, and iteratively refine image collection classifications entirely through language and intuitive interactions.

2.3 Semantic interaction

Semantic interaction (SI) techniques [1] allow systems to adapt dynamically to user feedback, often by learning from how users manipulate data visualizations. The ImageSI framework [8] extended this concept to 2D image projections, enabling users to change embedding spaces through drag-and-drop interaction. We don't use ImageSI's embedding manipulation pipeline, as we found that ImageSI performed poorly when repositioning images in cluster visualizations, particularly when dealing with more than two clusters. Therefore, we removed that component from our pipeline. The updates it produced were minimal and failed to reveal the distinctions we aimed to introduce. It remains unclear whether this underperformance was inherent to the method or a result of limitations in our reproduction. Had it worked as intended, this feature could have added a valuable layer of interaction to our dashboard, enabling more diverse and expressive user input.

Nevertheless, we adopt its concept of letting users adjust image descriptions interactively by drag and drop interactions, which encourages better understanding of how the system interprets visual content and provides a lightweight means to refine retrieval results.

2.4 User-Centered Visual Analytics

Our work also builds on principles from user-centered multimedia analytics [16, 17], emphasizing transparency, mixed-initiative interaction, and human-AI teaming. The ability to edit image descriptions supports alignment between user intent and model behaviour — a goal highlighted by recent work on interactive prompt refinement [2]. Furthermore, by providing per-image probabilities instead of only classifications, our system encourages exploration and insight, supporting a broader range of analytic tasks from casual browsing to focused dataset curation.

Our dashboard design is inspired by prior research in interactive multimedia analytics and visual retrieval systems. Rather than relying on static labels or metadata, our system centres on dynamic image retrieval driven by user-defined keywords and enriched with probability feedback.

Following the "3i" framework for interactive multimodal analysis [17], we integrate coordinated views of image content, text descriptions, and metadata. Users can issue natural language queries to retrieve relevant images. For each image, the system returns a probability score indicating the likelihood that it matches the query. This probabilistic feedback provides a nuanced view of relevance, allowing users to explore ambiguous cases and better understand the system's behaviour.

Guo et al. [2] present PromptMagician, an interactive environment that helps users iteratively refine text-to-image prompts through live previews, variant suggestions, and real-time feedback on prompt effectiveness. While PromptMagician focuses on optimizing generative outputs, its combination of interactive exploration and quantitative feedback inspired our decision to include a dedicated *Statistical Overview* alongside the 2D projection view in our system. In our case, the *Statistical Overview* provides dynamic summary statistics (e.g., total images, average probability, high-confidence counts) for the current dataset, helping users monitor and understand how their semantic prompts influence the classification and probability assessments of images, even as they interactively adjust and explore the projection space.

Our approach also draws on multimodal embedding techniques [4], combining visual features with textual metadata where available. This fusion allows users to formulate complex queries (e.g., "sunny beach photos taken in Spain") and supports richer explanations of why certain images are retrieved.

In summary, the dashboard is designed to support flexible, user-driven exploration of image collections, offering transparency through probabilities, adaptability through editable descriptions, and responsiveness through natural language interaction.

3 Methodology

We built an interactive image description editing dashboard that allows human feedback and semantic conditioning. This combination allows users to make modifications to the image descriptions via two different mechanisms: by directly instructing an LLM on how to modify their description, and also by dragging the images between the categories. These categories are related to the inserted keyword, and are split based on the match between each image description and the keyword: "Definite positive", "Uncertain", "Definite negative".

Our dashboard bridges computational logic with human intent. The entire pipeline is implemented in Python using the Dash framework, with a customized version of the Cytoscape component to support interactive drag-and-drop adjustments to the visualized categories.

At the core of our pipeline is the IC | TC framework [6], which classifies images based on natural language criteria. The criteria can vary from simple elements, such as "sunny" to complex terms such as "birds of prey". To this extent, IC | TC leverages both VLMs and LLMs. In our implementation of IC | TC, we make use of the GPT-4o-mini LLM, as well as its VLM ability, via API calls.

Before settling on GPT-4o-mini, we experimented with BLIP and LLaVA for the image description task. We compared the descriptions each model generated for a set of test images and found that GPT-4o-mini consistently produced the most accurate and faithful descriptions, closely matching the actual image content. Additionally, GPT-4o-mini adhered more reliably to our custom instructions, ensuring that all relevant aspects specified in the prompt were appropriately emphasized.

We specifically chose the mini version of GPT-4o to support real-time description modification, which is a key part of our system's interactive workflow. The lightweight nature of the mini model allows for faster response times and smoother user experience. However, for the initial generation of descriptions, performed as a preprocessing step in our implementation, it is also feasible to use the full GPT-4o model to produce even more accurate and detailed outputs.

To achieve the natural language classification of the images, two steps are followed. First, visual features of the images to-be-classified are extracted using the VLM. Image descriptions are generated based on the extracted features, together with a prompt to guide the description's focus points. For our purposes, we employ the following prompt for generating the image descriptions:

```
Analyze the following image and provide a rich,
well-structured description that covers:
Weather & climate: What are the atmospheric
conditions (sunny, overcast, rainy, hot, cold, etc.)?
People: Are there any people? If so, how many,
what are they doing, how are they dressed? If none,
explicitly note the absence.
Location: Where does this scene appear to take place
(city, countryside, indoors, landmark cues)?
Activity: What's happening in the scene-what actions
or events are taking place?
Please weave these details into a single, coherent
paragraph that gives the reader a vivid sense of the
image.
```

This initial step is done as pre-processing, and the generated descriptions of the images are pre-loaded into the app when launching it. As the second step, which assigns images to one of the three categories, using the keyword, the following prompt is given to the LLM, together with the image description:

```
You will be given a description of an image and one
or more filter criteria. Your task is to determine
how likely it is that the description entails all of
the given filters.
When you are given the filters and description, reply
with exactly:
{"entailment_probability": P}
Where P is a number between 0.0 and 1.0 indicating
your confidence that the image description matches
all filters. Do not include any additional text.
Prompt structure:
Filters: [FILTER1], [FILTER2], ...
Description:
[IMAGE DESCRIPTION GOES HERE]
Your filters for this instance are: [FILTER]
```

Next to the image collection, we also display a pie chart with some insight about the zones. The pie chart offers information about the amount of images and the percentage that fall into each category.

3.1 Adjusting the classification

To allow the user to tailor the classification to their liking, we enable two means of interacting with the system: a drag-and-drop functionality, and the option to make changes to the image descriptions. The drag-and-drop allows the user to move selected images between the displayed categories. By changing an image's description, the user overwrites the LLM's interpretation of the image using a text box.

In both cases, a request to update the image description is sent to the LLM using the following prompt:

```
You are an expert at modifying image descriptions
based on user prompts.
```

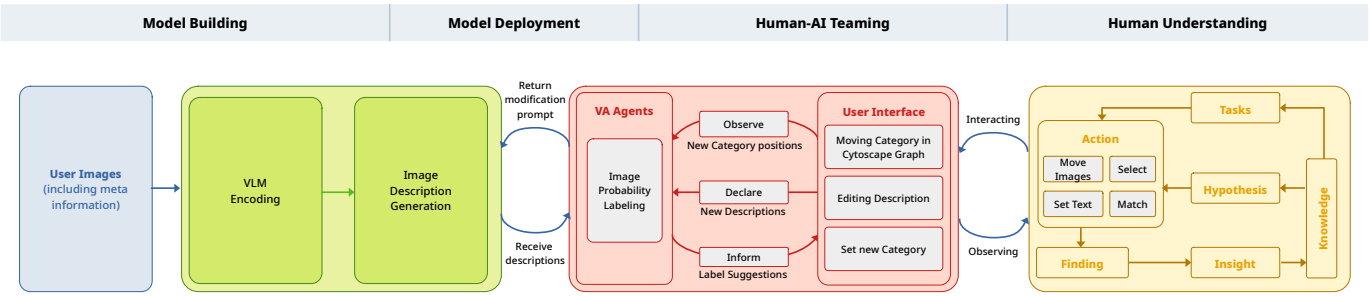

Fig. 2: Overview of the proposed multimedia analytics model. It extends upon and connects several existing models from visual analytics and multimedia analytics in a coherent, modern framework. Its focus lies in connecting emerging AI models with human understanding through a human-AI teaming component based on visual analytics agents and specialized visualizations. Here, the there are multiple feedback loops, such as new descriptions being generated based on the user guided labels or by direct description commands. Another loop is filter adjustment by the user such that new label probabilities are generated.

```
Your task is to take an existing image description
and modify it according to the user's instructions,
while maintaining the same style, detail level, and
structure as the original description.
Rules:
1.  Keep the same writing style and tone as the
original
2.  Maintain similar length and detail level
3.  Only change what the prompt specifically requests
4.  Preserve all other details that aren't
contradicted by the prompt
5.  Make the changes feel natural and integrated into
the description
Please modify the original description according to
the user prompt.  Make sure the changes integrate
naturally with the existing description while
preserving its style and detail level.
```

By incorporating the description modifications into an LLM prompt instead of simply appending the user input to the existing description, we ensure that prior erroneous elements from the description are removed, and that the user input fits well into the broader description. Moreover, it allows the user to provide the description modifications in natural language, instead of thinking about the desired wording of the description.

3.2 Keyword Filtering

Beyond manual adjustments, users can also input natural language descriptions to define how images should be grouped, for example: "travel photos with landmarks or scenery." These textual instructions are interpreted using the GPT-4o-mini LLM, which generates matching values between each image description and the keyword(s). The dashboard then updates the view to display the images in the category they have been assigned to.

All classification steps are executed server-side, ensuring that the interface remains lightweight and responsive even as computational tasks scale. The result is a flexible and adaptive system that blends automated feature learning with intuitive user control, enabling both exploratory and goal-directed analysis of image collections.

3.3 Dataset

Our system uses a subset of the *yffc100m* dataset [14]. This dataset consists of 100.000.000 media objects with metadata such as location, description, camera type etc. We took a random sample of 20 images for our purposes, and ensured that the random sampling did not introduce a strong bias in the distribution of locations or types of pictures.

4 Interaction Design

Our system builds upon and modifies the multimedia analytics model proposed by Worring et al. [16], emphasizing human-AI teaming through visual analytics agents and specialized visualizations, which can be seen in figure 2. We adjust this model by focusing on interaction mechanisms that support flexible, user-driven exploration of image collections, particularly when dealing with unique semantic details.

By enabling users to adjust descriptions of collections of images through natural language prompts and categorizing the collection based on keyword relevance, our system bridges high-level semantic guidance with low-level structural feedback. This interaction aligns with the insight generation framework by van Wijk [15], where both semantic and perceptual processes contribute to the construction of insight. Furthermore, our approach supports filtering strategies, as done in prior research [3, 10].

The main component of our dashboard is a 2D display of the image collection, *The image grid*, which is organized in categories. The user is able to enter keywords to determine the criterion used to categorize the images, based on the images' descriptions. Together with the drag and drop interaction, which also updates the description, there is a two-fold approach in adjusting the description. The intention is that, upon adjusting the groups and grouping the displayed pictures using keywords, the users can change description over multiple axes (the different key words).

The *Statistical overview* in the dashboard displays the size of the image categories as a pie chart, next to the quantitative display that the categorized image collection offers. Scrolling down on the app displays three tabs which show the isolated image thumbnails for each category. We choose to show the images in a basic grid, for efficiency [17]. Clicking on an image thumbnail allows the user to select the image. Upon doing so, a larger view of the image appears, along with other related information: the image metadata and file name is displayed, as well as its generated description and which category it belongs to.

By modifying the image description and/or the filter keyword, the user can reconfigure the view. Moreover, upon modifying the image description, the removed elements are displayed in red, and the newly added elements are presented in green font, offering an easy way for the user to understand the effect of their modifications.

4.1 Insight through Mixed-Level Interaction

Our system operationalizes the principles of Pike et al.'s Science of Interaction by promoting insight through a deliberate combination of cognitive (high-level) and perceptual (low-level) user engagements [10]. According to Pike et al., insight in visual analytics emerges through a dialogue between external visual representations and internal mental models. We translate this theory into practice by designing a mixed-level interaction framework that merges semantic reasoning with direct manipulation.

- **High-level interactions:** Users initiate the analytic process by defining natural language keywords, which serve as semantic criteria for image classification. This corresponds to the high-level reasoning in Pike's framework, where users' goals, knowledge, and hypotheses drive exploration. The *Classes tab* complements this process by allowing users to browse and assess retrieved results organized by match probability.
- **Low-level interactions:** Users refine system understanding through spatial manipulation—dragging images in the Cytoscape interface into one of three zones: (1) Definite Positive, (2) Uncertain, or (3) Definite Negative. These actions provide implicit feedback, instructing the model how to adjust the description. For example, dragging an image from the Definite Negative to the Definite Positive region signals that the image should be reinterpreted as a strong match for the given keyword. Multiple images can be adjusted simultaneously to support batch refinement. Another form of low-level interaction allows users to edit image descriptions directly using natural language prompts. These textual updates can relate to any attribute of the image, even if unrelated to the current keyword, offering a flexible mechanism to shape the semantic representation.

This dual-layer interaction design—semantic input coupled with perceptual adjustment—supports what Pike et al. describe as "analytic discourse," where mutual feedback between user intent and visual representations fosters hypothesis generation, verification, and iterative refinement.

5 Evaluation Design

Our evaluation follows best practices for human-centered evaluation of interactive machine learning systems, as outlined by Sperrle et al. [13]. The system is designed to support ongoing user evaluation during use, by surfacing insights into interaction effects and system responses.

Users can evaluate the system through three main interactions:

1. **Keyword-based retrieval and self-evaluation**, using the *Job Control Panel* and *Statistical Overview*.
2. **Manual refinement of categories**, through drag-and-drop interactions in the *Categorie Visualization*.
3. **Direct editing of image descriptions**, via natural language prompts in the *Image Selection and Editing*.

First, the system enables users to retrieve images based on keyword matching via the *Job Control Panel*. Users can input one or more keywords, or entire sentences, to which the retrieved images should correspond. To assess how many images match the specified criteria, users can consult the *Statistical Overview*, which presents summary analytics in the form of a pie chart. The chart displays three categories: images that match, do not match, or may match the keywords. This visualization provides insights into the distribution of matching probabilities and helps users reflect on the quality and specificity of their keyword prompts. For example, a high proportion of uncertain matches may indicate vague or contradictory keywords.

Second, the system allows users to refine the categories by dragging images within the *Categorie Visualization*, adjusting their position to reflect a more accurate match probability (see Section 6 for details). Upon submitting these adjustments, the system updates the image descriptions accordingly, enabling users to observe how the model adapts to manual corrections. This interaction supports evaluation of the system's responsiveness to drag-and-drop manipulations.

Third, users can directly edit image descriptions through natural language prompts (also detailed in Section 6). After submitting changes, users can inspect both the updated description and the resulting categories' placement, providing feedback on how well the model integrates user input. The new description should appropriately reflect the user's modifications.

By enabling users to iteratively retrieve, adjust, and re-evaluate image collections through keyword queries and interactive editing, the system maintains a continuous human-in-the-loop process. This supports mixed-level interaction [10], where users alternate between semantic input (text queries) and perceptual feedback (visual classification), fostering deeper insight and understanding as recommended by Sperrle et al. [13].

Due to time constraints, we were not able to have the system evaluated by a professional designer or a researcher specializing in visual analytics or interaction design. While our internal testing and user walkthroughs provided valuable insights into usability and interaction flow, a more formal evaluation by domain experts would be necessary to rigorously assess the design quality and generalizability of the interaction techniques.

In summary, the system design facilitates ongoing user-driven evaluation during interaction, addressing key dimensions of human-centered evaluation of interactive machine learning systems, including trust, interpretability, co-adaptation, and human agency.

6 Implementation Design

Our implementation is built upon the Dash/Plotly framework, using interactive components to support human-in-the-loop labelling of image datasets. The foundational design was designed in a modular way, with extensions to support multimodal image clustering and manipulation.

6.1 Layout and User Interface

The interface employs a grid-based layout, presenting users with multiple coordinated views and controls. Key components include:

- **Job Control Panel** (1 in Figure 1): Users can submit text-based filter prompts, which are used to retrieve and cluster relevant image collections. This is implemented in the *JobPanel* widget, which interacts with a SLURM job manager backend to trigger clustering jobs and refresh the dataset.
- **Statistical Overview**(2 in Figure 1): The *StatCards* widget provides summary statistics (e.g., total images, average probability, high-confidence counts) for the current dataset, updating dynamically as the data changes.
- **Category Visualization**(3 and 4 in Figure 1): The *ChartPanel* combines a 2D scatter plot (using Plotly) and a cluster distribution pie chart, giving users an overview of the clustering results and confidence levels. The scatter plot encodes cluster membership and probability, with interactive selection capabilities.
- **Image Grid**(6 in Figure 1): The *ImageGrid* widget displays images grouped by category, allowing users to browse and select images within each cluster.
- **Image Selection and Editing**(5 in Figure 1): The *SelectedImage* widget enables users to select an image (via clicking in the scatter plot or grid) and modify its description using a text prompt. Modifications are processed by a backend script (`modify_description.py`) that uses GPT-4o-mini to generate new descriptions.

6.2 Interactive Clustering and Editing

Users can enter natural language prompts to filter and cluster the image dataset. This interaction is inspired by the approach of Kwon et al. [6], and is implemented by updating a job file and submitting it to the backend for processing. The results are loaded and visualized in real time.

Beyond passive exploration, the interface supports active modification of clusters. In the 2D scatter plot (implemented with Dash Cytoscape in the *ImageGraph* widget), users can drag and drop images to new positions. Upon submitting changes, the system updates the cluster assignments based on the new positions, and triggers the generation of new image descriptions reflecting the updated cluster context.

Besides this interaction, for any selected image, users can provide a custom prompt to modify its description. Here the backend script also uses the OpenAI API to generate a revised description, which is then saved and reflected in the dataset.

Cluster locations within the 2D scatter plot are initialized randomly within their respective sections, primarily to maximize use of screen space and avoid visual clutter. This design choice is motivated by the findings of Kucher et al. [5], who argue that two-dimensional displays are generally more intuitive and easier to interpret for users exploring complex datasets

7 Limitations & Future work

Our current system provides a foundation for interactive text-to-image retrieval with editable semantic descriptions and probabilistic feedback. However, several limitations and opportunities for future development remain.

At present, our system is designed for small-scale datasets due to computational constraints. Scaling the pipeline to handle larger image collections would be interesting. Additionally, the lack of formal performance metrics limits the ability to quantify system effectiveness. Interesting would be to incorporate evaluation metrics such as Best Log Rank Integral (BRI) [7], which are specifically designed for interactive retrieval systems, to enable more systematic and reproducible benchmarking.

Our image descriptions are generated based on predefined prompts focusing on specific visual aspects (e.g., weather, people, activity). While this approach improves consistency, it makes it less robust. How to handle this is an interesting topic.

Due to time constraints, we were not able to involve professional designers or visual analytics researchers in the evaluation process. A formal design study involving these stakeholders would help assess the system's usability, and effectiveness of its visual encodings.

While we experimented with the ImageSI framework to enable embedding manipulation via drag-and-drop, its practical utility was limited. Future work could explore more expressive and learnable forms of semantic interaction.

Finally, inspired by systems like PlugIR [7], we see potential in integrating dialogue-based interfaces that allow the system to actively ask follow-up questions when user queries are ambiguous or not well specified. This would reduce noisiness, improve retrieval precision, and make the interaction more collaborative.

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